A Deep Transfer Learning Method for Generating High Spatiotemporal Resolution Leaf Area Index From Landsat Surface Reflectance

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***Abstract*—Leaf area index (LAI) is an important biophysical parameter characterizing vegetation structure and function in ecosystem modeling and climate studies. While medium-resolution LAI products have been widely used for global monitoring, they cannot adequately capture fine-scale heterogeneity. High-resolution satellite data provide new opportunities for LAI estimation, but two major challenges remain: 1) significant discrepancies between satellite-derived and ground-measured LAI due to reliance on satellite products or simulation data for model training; and 2) time-series discontinuity caused by cloud contamination. To address these challenges, this study proposes a deep transfer learning framework to generate seamless high spatiotemporal resolution LAI. The framework consists of three steps: First, a deep belief network (DBN) model is pretrained using coarse-resolution satellite products to establish universal spectral-LAI relationships. Second, the model is fine-tuned with global ground measurements from ImagineS and ground-based observation validation (GBOV) networks, effectively bridging the gap between satellite retrievals and field observations. Third, a light gradient boosting machine (LightGBM)-based reconstruction model integrating spatiotemporal constraints and environmental variables fills cloud-induced gaps to achieve continuous LAI time series. Comprehensive validation using field measurements across seven land cover types from 2013 to 2021 demonstrates significant accuracy improvements (overall root-mean-square error (RMSE) = 1.06–1.11), with particularly notable enhancements in high-LAI forests (RMSE reduced from 1.47–1.67 to 0.81–0.95). Compared with existing spatiotemporal reconstruction methods, our approach achieves superior spatial detail preservation and temporal continuity. As a demonstration, we generated a seamless 30-m, eight-day LAI product for China spanning 2013–2021. This study highlights the value of deep transfer learning for cross-scale vegetation parameter retrieval by combining large-scale satellite pretraining with limited ground observations to generate high-quality LAI products.**

***Index Terms*—Deep belief network (DBN), high spatiotemporal resolution, leaf area index (LAI), light gradient boosting machine (LightGBM), transfer learning.**

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I. Introduction

LEAF area index (LAI) represents a fundamental biophysical metric quantifying the total one-sided leaf surface area per unit ground area, serving as a critical indicator of vegetation canopy architecture and physiological functioning [1]. This parameter governs the bidirectional transfer of energy, moisture, CO_2, and various biogeochemical fluxes across the terrestrial-atmospheric interface [2]. Continuous LAI monitoring enables the detection of vegetation dynamics responding to climatic variations, anthropogenic land modifications, and diverse environmental pressures [3]. Furthermore, serving as an essential input variable for ecosystem modeling frameworks, LAI facilitates the simulation of plant developmental processes, surface-atmosphere coupling, and ecological functioning across multiple spatial and temporal dimensions.

As satellite remote sensing technologies have evolved, numerous intermediate-resolution (250 m–1 km) global LAI datasets have been developed and extensively utilized, including the Moderate-Resolution Imaging Spectroradiometer (MODIS) [4], Visible Infrared Imaging Radiometer Suite (VIIRS) [5], and Global Land Surface Satellite (GLASS) [6]. These products achieve consistent spatial coverage and temporal repeatability, driving research in global vegetation mapping, terrestrial carbon cycle analysis, and benchmarking of ecosystem models [7]. However, the medium resolution of these products presents limitations for applications that require or benefit from fine-scale characterization of vegetation heterogeneity. In fragmented or complex canopies, rough

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pixel sizes tend to mask fine-scale variability, thus hindering ecological studies that need to distinguish between adjacent vegetation types, gradients, and patches [8]. In addition, moderate resolution limits the utility of precision agriculture that focuses on field variation. Therefore, new approaches with higher resolution LAI are needed to advance research and applications in vegetation science, carbon cycle analysis, ecosystem modeling, and precision agriculture.

The Landsat satellite constellation offers distinct benefits including open data access, fine spatial resolution (30 m), extensive geographic coverage, comprehensive spectral bands, and decades-long observational records, establishing it as a premier resource for high-resolution LAI estimation across multiple temporal scales. While Sentinel-2 provides comparable spatial resolution (10–20 m) with higher temporal frequency (five-day revisit), longer historical archive (since 1972) and well-established radiometric consistency across missions of Landsat make it particularly suitable for developing robust global models and enabling multidecadal vegetation monitoring. The fine granularity of Landsat pixels can capture vegetation heterogeneity, while extensive databases can reconstruct dense time series. However, obtaining continuous LAI time series from Landsat data is challenging due to few clear observations, cloud cover, and 16-day revisit periods. Existing high-resolution LAI retrieval methods can be roughly divided into three categories: physical model-based methods, data-driven methods, and hybrid techniques.

Physical model-based methods, which mainly rely on radiative transfer models (RTMs), provide a solid theoretical basis for LAI estimation [9]. These methods model the interaction between solar radiation and vegetation canopy, allowing the inversion of biophysical parameters from observed reflectance data. For example, the PROSPECT-SAIL (PROSAIL) model has been widely used to retrieve LAI from various satellite sensors, including Landsat [10]. The advantage of the physical model is that it is based on the well-known theory of radiative transport, which makes it possible for it to be applicable to different ecosystems [11]. However, they face significant challenges when applied to large areas of high-resolution data. The complexity of RTM often leads to ill-posed inversion problems, where multiple parameter combinations can produce similar reflection features [12]. In addition, at high resolution with significant intrapixel heterogeneity, many canopy uniformity assumptions of RTM are problematic. The computational intensity of RTM inversion also poses a major barrier to large-scale applications, often requiring simplified or look-up table methods that can compromise accuracy [13].

Data-driven methodologies, encompassing machine learning and deep learning paradigms, have gained popularity due to their capacity to model intricate nonlinear associations between satellite observations and vegetation parameters without requiring explicit physical formulations. Random forests, support vector machines, and neural networks have been successfully applied to estimate LAI from Landsat data [14], [15]. These approaches can process high-dimensional input datasets effectively and demonstrate strong capabilities in representing spatial heterogeneity. Advanced deep learning architectures, including convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, have propelled the discipline forward through automated extraction of hierarchical patterns and temporal dependencies within satellite time series [16], [17], [18]. However, the performance of data-driven methods depends heavily on the quality and representation of the training data. At high resolution, obtaining sufficient ground truth data for model training and validation becomes a major challenge. In addition, these methods often struggle with extrapolations beyond the scope of the training data, which can lead to unreliable estimates under new or extreme conditions [19]. The lack of clear physical constraints can also lead to biophysically implausible predictions that require careful postprocessing and quality control.

Hybrid technologies are designed to combine the benefits of physical models and data-driven approaches [20]. These methods typically use RTM to generate comprehensive training data for machine learning algorithms, allowing for the merging of physical constraints while leveraging the efficiency and flexibility of data-driven approaches [21]. For example, Bayesian methods have been used to combine prior knowledge of RTM with empirical observations in a probabilistic framework [22]. Another hybrid strategy involves using machine learning to simulate complex RTM, significantly reducing computational costs while maintaining physical consistency [23]. While promising, hybrid approaches face challenges in balancing physical realism and empirical accuracy [24]. Generating realistic synthetic data that adequately represents the variability and complexity of real-world conditions remains a key challenge. Furthermore, the propagation of uncertainty from physical models to data-driven components requires careful consideration to ensure reliable quantification of uncertainty in the final product [25].

Despite great advances in these methods, generating long-term, high-precision, and gap-free LAI time series at Landsat resolution over a large range remains challenging. A main issue is how to reconstruct the missing LAI values caused by clouds. Commonly used spatial–temporal fusion methods, data assimilation methods, sequential deep learning methods, etc., are generally unable to effectively solve the problem of missing Landsat LAI reconstruction in cloudy and foggy areas, and these methods often require a large number of high-quality observations to obtain relatively stable performance [26]. In particular, the availability of raw Landsat data during vegetation phenological transition and peak growth period largely determines the reliability of the method [27]. One possible approach is to integrate easily accessible climate and environment variables and coarse-resolution product experience values to drive a machine learning regression model to predict missing values. This method has been widely used in the production of land surface parameters such as fractional forest cover (FFC) [28], land surface temperature (LST) [29], soil moisture [30], and so on. Nevertheless, this method still lacks the ability to capture spatiotemporal information, making it susceptible to the noise of individual pixels during a given period [31]. If the constraints of temporal and spatial information can be included, it may show a good effect in producing high-quality Landsat LAI products [32].

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A secondary challenge concerns the discrepancy between field-measured LAI and satellite-derived LAI, which constrains the precision of LAI retrieval algorithms, primarily attributed to scale disparities between ground-based LAI observations and satellite LAI estimates [33]. Some studies have tried to combine Landsat surface reflectance data with coarse-resolution LAI products such as GLASS or MODIS to generate high-resolution LAI using machine learning methods [14]. Although they will try to screen homogeneous sample points to ensure the generalization ability of the model, this model is strictly suitable for coarse scale. Therefore, the saturation effect of Landsat LAI could not be effectively reduced. Integrated deep learning frameworks that merge ground-based LAI observations with existing coarse-resolution LAI datasets represent a promising solution to address this challenge and can effectively bridge the discrepancy between satellite-derived and field-measured LAI measurements [34]. However, compared with RTM simulation or product-based training samples, the number of measured LAI values is too sparse, and the unbalanced samples of these combinations may lead to the deep learning model ignoring the characteristics of ground-measured LAI. To address this challenge, deep transfer learning can be introduced to integrate different knowledge from satellite products and ground-measured values [35], thereby transferring satellite-based pretrained inversion models to domains applicable to measured LAI values and mitigating the uncertainty caused by differences between the two.

In this study, we propose a deep transfer learning framework using deep belief networks (DBNs) to address the above challenges and generate a seamless 30-m, eight-day LAI product. DBN was selected for the pretraining stage due to its distinctive advantages in our framework. Unlike CNN-based approaches that require spatial context windows, DBN performs point-to-point mapping using only spectral information from individual pixels, making it particularly suitable for learning the relationship between MODIS surface reflectance and GLASS LAI. Compared to LSTM architectures that necessitate complete temporal sequences as input, DBN can process individual time steps independently, avoiding complications from cloud-induced gaps in satellite time series during pretraining. Furthermore, the layer-wise unsupervised pretraining mechanism of DBN enables efficient hierarchical feature learning from large-scale datasets, establishing robust spectral-LAI representations that transfer effectively across spatial scales from coarse-resolution satellite products to fine-resolution ground measurements. First, we developed a pretrained DBN model for LAI inversion based on MODIS land surface reflectance and the GLASS LAI product. Second, transfer learning technology was applied to incorporate ground-measured LAI data to fine-tune the pretrained DBN model, and the fine-tuned model was used to estimate Landsat clear-sky LAI. Finally, a missing LAI reconstruction light gradient boosting machine (LightGBM) model was constructed to estimate Landsat cloudy-sky LAI by integrating climate, environment, coarse-resolution LAI products, and Landsat spatiotemporal information. We conducted a comprehensive evaluation of the effectiveness of our approach using globally in situ LAI measurements and other advanced LAI products. Compared to existing high-resolution products such as Sentinel-2 LAI and GLASS Landsat LAI, our framework offers three key improvements: mitigating high-value saturation through ground-measurement calibration, maintaining spatiotemporal continuity under prolonged cloud cover (>30 days), and achieving superior accuracy across diverse biomes.

TABLE I
LANDSAT 8 OLI BAND INFORMATION USED IN THE STUDY

Band	2	3	4	5	6	7
Name	Blue	Green	Red	NIR	SWIR1	SWIR2
Width(μm)	0.43-0.51	0.53-0.59	0.64-0.67	0.85-0.88	1.57-1.65	2.11-2.29
Resolution(m)	30	30	30	30	30	30

II. DATASETS

A. Landsat Data

Landsat 8 surface reflectance product from 2013 to 2021 is one of the main input data for the deep transfer learning model in this study. Launched in 2013 as part of the long-term Landsat program, Landsat 8 carries an Operational Land Imager (OLI) sensor that provides 30-m resolution multispectral images over a 16-day revisit cycle [36]. We focused on six spectral bands: blue (0.452–0.512 μm), green (0.533–0.590 μm), red (0.636–0.673 μm), near infrared (0.851–0.879 μm), and two shortwave infrared bands (1.566–1.651 μm and 2.107–2.294 μm). Top-of-atmosphere reflectance values are computed through established radiometric calibration protocols, whereas surface reflectance datasets are processed via the Landsat Surface Reflectance Code (LaSRC) methodology, which corrects for atmospheric interference, solar illumination conditions, and viewing geometry parameters [37]. This atmospheric correction ensures consistent physical values across time and space, which is essential for reliable biophysical parameter retrieval. To identify clear-sky observations, we adopt the Fmask 4.6 algorithm, which provides pixel-level clouds and cloud shadow masks [38]. Table I gives details of the Landsat 8 OLI band selected for this study, including its spectral range and spatial resolution.

B. GLASS and MODIS LAI Product

The GLASS Version 6 LAI dataset delivers comprehensive temporal coverage of LAI across global territories [6]. This dataset features 250-m spatial resolution with eight-day temporal intervals, encompassing the period from 2000 to 2021. The dataset employs a sophisticated bidirectional LSTM (Bi-LSTM) deep learning architecture trained on MODIS surface reflectance observations. Within our research framework, the GLASS V6 LAI product was dual-purpose. First, it provides valuable learning knowledge for the DBN pretraining model of Landsat LAI estimation in clear sky, which is helpful to constrain parameter space and improve the accuracy of LAI retrieval. Second, it functions as a benchmark dataset for comparative assessment of our developed 30-m Landsat all-sky LAI products.

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The MODIS Collection 6 MCD15A2H dataset delivers global LAI estimations at 500-m resolution through eight-day composite intervals [39]. This dataset was generated using a radiative transfer-based algorithm applied to MODIS surface reflectance data [40]. Comprehensive validation studies demonstrate that MODIS LAI products successfully characterize seasonal and annual vegetation fluctuations across various ecosystem types. However, some studies have pointed out that it may be underestimated in dense canopies due to signal saturation [41]. In our study, the MODIS MCD15A2H LAI product serves as an independent validation dataset for assessing our 30-m Landsat LAI retrievals.

C. DEM and Reanalysis Data

Terrain and environmental variable data are important input features driving the reconstruction model of missing LAI. This investigation employs topographical characteristics including 30-m digital elevation models (DEMs), terrain slope, and aspect data derived from the Shuttle Radar Topography Mission (SRTM) to represent landscape features influencing LAI spatial patterns. Environmental forcing variables from European Centre for Medium-Range Weather Forecasts (ECMWF) Fifth-Generation Reanalysis (ERA5) dataset support our gap-filling LAI reconstruction approach. ERA5 represents the most recent global atmospheric reanalysis product developed by the ECMWF, delivering continuous environmental records from 1979 onward [42]. We selected three key variables: air temperature, incident shortwave radiation, and vapor pressure deficit data [43]. The original dataset maintains $0.25^\circ \times 0.25^\circ$ spatial resolution with hourly temporal granularity. To match Landsat observations, we resampled the ERA5 data to 30 m using bilinear interpolation, which is appropriate for continuous environmental variables that exhibit smooth spatial gradients at the 30-m scale, and extracted the environmental variable values that most closely matched Landsat transit times. While this resampling introduces spatial smoothing, the environmental drivers typically vary gradually across local scales, making ERA5-derived values physically meaningful for characterizing general environmental conditions at individual Landsat pixels.

D. In Situ LAI Measurements

In situ LAI measurements from two main observation networks were used to evaluate our approach, including ImagineS and ground-based observation validation (GBOV), which together provide multiple validation data for different biomes and climates. The ImagineS network covers ecosystems from the north to the tropics in Europe, Africa, and South America [44]. LAI measurements were taken from digital hemispherical photographs using standardized image processing techniques. These data have been widely used to evaluate various satellite derivatives, including MODIS, GEOLAND2 Version 1 (GEOV1), and SPOT/VGT [45]. The GBOV network is part of the Copernicus program and integrates ground LAI from more than 30 sites worldwide [46]. Using standardized protocols, GBOV integrates multiple observation methods to provide consistent reference data across different biomes for validation of Sentinel-2 and 3 satellite products. Collectively, these two networks provide extensive, multiyear terrestrial measurements of LAI, and the diversity of ecosystems represented in these datasets, from the Mediterranean to the Arctic biome, allows satellite-derived LAI products to be fully validated at different spatiotemporal scales.

We implemented quality control procedures for measured LAI data to improve the reliability of the validation process. We plotted measured LAI against the corresponding Landsat normalized difference vegetation index (NDVI) to visualize trends and identify potential outliers. The data pairs were then grouped into boxes with an NDVI interval of 0.05. In each box, we applied a statistical filter: samples with NDVI values falling below the 10th percentile or exceeding the 90th percentile were identified as potentially erroneous and removed from subsequent analysis [14], [47].

The ImagineS project provides 30-m resolution LAI reference maps for direct comparison with our Landsat LAI; this allows us to perform spatial evaluations of our estimates in different landscapes. For larger scale validation, we used the Direct2.1 dataset, which provides an extensive collection of LAI measurements at the 3-km scale [48]. These coarser resolution data are used to assess the consistency of our Landsat LAI when aggregated to 3 km and compare their performance to MODIS and GLASS LAI at the same scale.

III. Methodology

Our spatiotemporal seamless high-resolution LAI estimation framework based on deep transfer learning includes three primary steps, illustrated in Fig. 1. First, we develop a DBN model for clear-sky LAI retrieval by pretraining with MODIS surface reflectance inputs and GLASS V6 LAI targets. Second, we employ Landsat surface reflectance observations combined with field-measured LAI data to refine the pre-established DBN model, applying the optimized model to derive clear-sky LAI estimates from Landsat 8 surface reflectance. Third, we integrate spatiotemporal constraint information, prior knowledge, and environmental variables to construct a new LightGBM model for gap-filling Landsat LAI under cloudy conditions.

A. Pretraining a DBN Model for Clear-Sky LAI Estimation

DBNs show superior performance in learning complex nonlinear relationships from remote sensing data [49]. Compared with traditional machine learning methods, DBN can automatically learn the hierarchical feature representation of data through its multilayer structure, which makes it particularly suitable for extracting LAI-related feature patterns from MODIS surface reflectance data. The DBN consists of multiple constrained Boltzmann machine (RBM) layers and a backpropagation (BP) network [50]. RBM is an energy-based probabilistic model in which intralayer neurons are independent of each other, helping to efficiently learn the probability distribution of input data [51]. When multiple RBMS are stacked on top of each other, the lower layers can learn basic features (such as spectral features), while the higher layers can capture more abstract features (such as vegetation structure

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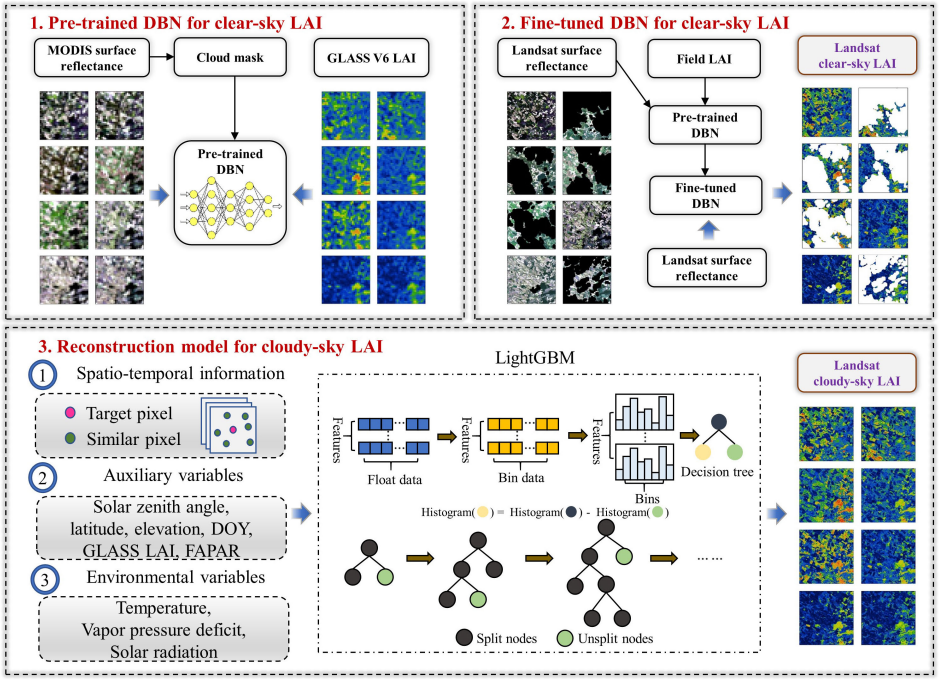

Fig. 1. Flowchart for generation of high spatiotemporal resolution LAI using deep transfer learning.

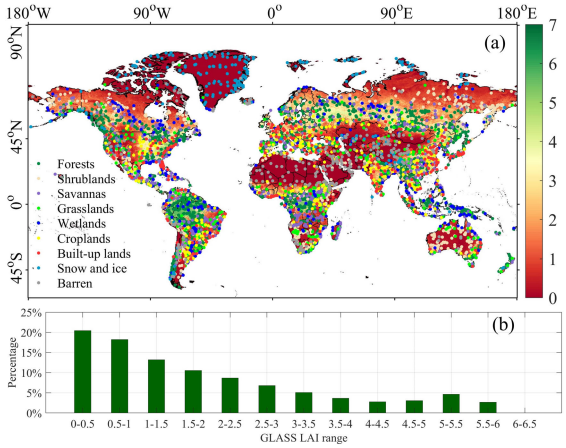

Fig. 2. (a) Global distribution of representative pixels used for training the deep transfer learning model, and the base map shows the GLASS V6 250-m LAI product for day 233 of 2021. (b) Distribution of LAI ranges for these pixels.

information). In our network structure, four layers of RBM are stacked in sequence, with the number of neurons [512, 256, 128, 64]. The lower layer learns the basic spectral features, while the higher layer captures the more abstract vegetation structural features. The network input is MODIS land surface reflectance data, and features are extracted through layer-by-layer unsupervised pretraining. Finally, features are mapped to target LAI values by a two-layer [64, 32] BP network. In order to enhance the nonlinear expression ability and generalization performance of the model, a dropout mechanism (dropout rate 0.3) is introduced between RBM layers, and tanh is used as the activation function. In addition, the mean squared error (mse) is used as a loss function to optimize the model parameters.

Our methodology incorporates the representative pixels used for GLASS V6 LAI product development (Fig. 2). These pixels, totaling 52 999 globally distributed samples, were obtained through analysis identifying LAI temporal sequences exhibiting minimal discrepancies among GLASS V5, MODIS, and PROBA-V V1 LAI datasets. The global distribution of representative pixels is available in [32]. We extracted GLASS V6 LAI and MODIS surface reflectance corresponding to these pixels to construct the pretraining samples. The model training adopts a two-stage strategy. First, we implement stochastic gradient descent (SGD) optimization to sequentially pretrain each RBM layer [52]. The initial learning rate is configured at 0.01 with cosine annealing scheduling, while the k value of the contrast divergence algorithm was set to 1. Then, the BP network will conduct a maximum of 200 rounds of fine-tuning training; the batch size was set to 256. We incorporate early termination protocols, halting training when validation loss fails to improve across five consecutive epochs. The entire training process covers all periods from year-day (DOY) 1 to 365, with a time interval of eight days.

B. Fine Tuning of DBN Model

Due to various influencing factors such as scale effect, saturation effect, canopy clustering effect, local atmospheric and soil background conditions, there are often significant differences between LAI based on satellite inversion and ground measurements [33]. While our pre-established DBN architecture has acquired generalized nonlinear associations linking surface reflectance with LAI, the model may not be able to fully adapt to the feature distribution of ground measurements. Consequently, we implement transfer learning methodologies to refine our pre-established DBN framework,

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addressing disparities between satellite-based LAI products and field-measured values [53].

In the fine-tuning phase, we adopt an efficient transfer learning strategy: freeze the parameters of the RBM layer in the pretrained DBN model and fine-tune only the BP layer. The RBM layer has learned the general feature representation of the surface reflectance data during the pretraining process, while the BP layer is responsible for mapping these features to the final LAI estimates. By fine-tuning the BP layer, the ability to extract basic features can be maintained, while the mapping relationship between features and LAI can be optimized to better adapt to the ground-measured LAI. We fine-tuned the pretrained model using high-quality ground LAI measurements from a global network of ImagineS and GBOV that cover different vegetation types and environmental conditions. These 40 sites were randomly divided into five groups with eight sites in each group. In each iteration, measured LAI data from four groups (32 sites) served for model refinement, while data from the remaining one group (eight sites) provided independent assessment.

C. Reconstruction of Cloudy-Sky LAI Using Spatiotemporal Information and Environment Variables

LightGBM model was used to reconstruct LAI on cloudy days, which is an efficient gradient-boosting decision tree (GBDT) algorithm developed by Microsoft. The main reason for choosing LightGBM is its excellent computational efficiency and prediction accuracy in processing large-scale spatial data, which is particularly suitable for capturing complex nonlinear relationships between LAI and environmental variables. Contrasting with conventional GBDT approaches, LightGBM substantially minimizes memory requirements and accelerates training through gradient one-side sampling (GOSS) and exclusive feature bundling (EFB) methodologies, proving essential for processing our comprehensive Landsat spatiotemporal collections [54].

LightGBM adopts an iterative approach for constructing decision tree ensembles, where each successive tree enhances predictions by modeling residuals from preceding trees, progressively approaching actual LAI values through loss function minimization. For model performance optimization, we utilized RandomizedSearchCV for hyperparameter calibration, assessing various parameter combinations via fivefold cross-validation [55]. The search space included: n_estimators [100, 300, 500, 700], learning_rate [0.01, 0.05, 0.1, 0.2], max_depth [5, 7, 9, 11], min_child_samples [50, 100, 150, 200], num_leaves [31, 63, 121, 255], reg_alpha [0, 0.1, 0.5], and reg_lambda [0, 1, 5, 10], with 100 iterations performed. After system tuning, the optimal parameter configuration is finally determined as follows: 500 estimators, 0.1 learning rate, maximum depth of 7, minimum leaf sample weight of 100, maximum leaf count of 121, L1 regularization of 0.1, and L2 regularization of 5. The global representative pixels described in Section III-A were also used to develop the LightGBM Landsat cloud-sky LAI reconstruction model. To evaluate our approach, gap filling and Savitzky–Golay filter (GFSG) [56], STAIR [57], Fit-FC [58], flexible spatiotemporal data fusion (FSDAF) [59], enhanced FSDAF (SFSDAF) [60], spatial and temporal adaptive reflection fusion model (STARFM) [61], spatial and temporal nonlocal filter-based data fusion method (STNLFFM) [62], bidirectional temporal convolutional network (BiTCN), and Transformer were also applied to the temporal reconstruction of Landsat LAI.

The input variables of the LightGBM model consist of the following parts. The first is spatiotemporal information. For clear-sky Landsat LAI (Lt) on the target date, we first extracted Landsat LAI values for the same date of the previous year and the following year (Ln1 and Ln2). These two values represent the time features most associated with the target Landsat LAI values. Next, based on the seasonal changes of Landsat LAI, a spatial window of 3×3 km (100×100 Landsat pixels) was adopted to identify similar pixels. This window size was determined through sensitivity analysis using all validation sites listed in Table II. Testing five window sizes (1, 2, 3, 5, and 10 km) showed that reconstruction root mse (RMSE) improved from 1.44 (1 km) and 1.25 (2 km) to 1.11 (3 km), then stabilized at 1.09 (5 km) and 1.08 (10 km). While larger windows yielded marginally lower RMSE, the 3-km window provides optimal balance between reconstruction accuracy and computational efficiency, as windows exceeding 5-km substantially increase processing time without significant accuracy gains. We constructed LAI seasonal curves for all pixels in the spatial window by calculating the average eight-day LAI values for three years (current year, previous year, and next year). Seasonal curves are considered valid if at least 70% of LAI values (i.e., 33 of the 46 values) are available; if this threshold is not met, we extend the time frame for constructing the seasonal curve to five years, and so on, until the 70% criterion is met. This adaptive temporal window balances the need for robust phenological characterization with temporal relevance. In regions experiencing abrupt disturbances (e.g., fires, deforestation), the correlation-based similar pixel selection naturally filters out pixels with dramatically altered seasonal patterns, as disturbed pixels exhibit low correlation coefficients with stable vegetation trajectories. By calculating the correlation coefficient between the seasonal curve of the target pixel and other pixels in the spatial window, the first 3 pixels with the highest correlation coefficient were selected to determine the similar pixels. For each similar pixel, we extracted nine Landsat LAI values (Ln3–Ln11) for the current, previous, and subsequent years on the same date. These nine Landsat LAI values represent secondary spatiotemporal features associated with the target Landsat LAI. The spatiotemporal information represented by Ln1–Ln11 was used as the first 11 input features to drive the Landsat cloud LAI reconstruction model.

The second set of input features (Ln12–Ln16) includes solar zenith angle, latitude, altitude, day of the year (DOY), and GLASS LAI product values. The GLASS coarse-resolution values provide crucial prior knowledge; theoretically, comparable associations should exist between analogous high-resolution pixels and corresponding coarse-resolution counterparts. It is important to note that incorporating 250-m GLASS LAI as a feature for 30-m Landsat LAI reconstruction introduces spatial scale mismatch. However, GLASS LAI

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TABLE II
BASIC INFORMATION OF THE USED VALIDATION SITES

Sites	Network	Latitude	Longitude	Land Cover	Year	Number of samples
Pshenichne	ImagineS	50.0766	30.2322	Croplands	2013-2015	312
Collelongo	ImagineS	41.8500	13.5900	Deciduous Broadleaf Forest	2015	39
SouthWest1	ImagineS	43.5511	1.0889	Croplands	2013	60
SouthWest2	ImagineS	43.4471	1.1450	Croplands	2013	65
25 de Mayo	ImagineS	-37.9254	- 67.8038	Croplands	2014	45
Barrax	ImagineS	39.0544	-2.1007	Croplands	2014-2015	76
San Fernando	ImagineS	-34.7233	- 71.0015	Croplands	2015	40
CRE	ImagineS	43.9936	-0.0469	Croplands	2015	8
URG	ImagineS	43.6397	-0.4340	Croplands	2015	7
MTO	ImagineS	43.5728	1.3745	Croplands	2015	2
PEY	ImagineS	43.6662	0.2196	Croplands	2015	12
CON	ImagineS	43.9743	0.3360	Croplands	2015	8
SAV	ImagineS	43.8242	1.1749	Croplands	2015	10
LaReina1	ImagineS	37.8189	-4.8624	Croplands	2014	23
LaReina2	ImagineS	37.7929	-4.8267	Croplands	2014	26
Capitanata	ImagineS	41.4637	15.4867	Croplands	2014-2015	248
Maragua	ImagineS	-0.7720	36.9742	Croplands	2016	24
BART	GBOV	44.0639	-71.2873	Mixed Forest	2014-2021	225
BLAN	GBOV	39.0603	-78.0716	Deciduous Broadleaf Forest	2015-2021	154
CPER	GBOV	40.8155	-104.7460	Grasslands	2014-2021	198
DELA	GBOV	32.5417	-87.8039	Deciduous Broadleaf Forest	2016-2021	127
DSNY	GBOV	28.1250	-81.4362	Open Shrublands	2013-2021	349
GUAN	GBOV	17.9696	-66.8687	Evergreen Broadleaf Forest	2015-2021	619
HAIN	GBOV	51.0785	10.4506	Mixed Forest	2019-2021	266
HARV	GBOV	42.5369	-72.1727	Mixed Forest	2014-2021	292
JERC	GBOV	31.1948	-84.4686	Evergreen Needleleaf Forest	2013-2021	778
NIWO	GBOV	40.0543	-105.5820	Evergreen Needleleaf Forest	2015-2021	11
ONAQ	GBOV	40.1776	-112.4520	Open Shrublands	2014-2021	7
ORNL	GBOV	35.9641	-84.2826	Mixed Forest	2014-2021	316
OSBS	GBOV	29.6893	-81.9934	Evergreen Needleleaf Forest	2013-2021	376
SCBI	GBOV	38.8929	-78.1395	Mixed Forest	2014-2021	215
SERC	GBOV	38.8901	-76.5600	Croplands	2015-2021	247
STEI	GBOV	45.5089	-89.5864	Deciduous Broadleaf Forest	2015-2021	178
STER	GBOV	40.4619	-103.0290	Grasslands	2014-2021	19
TALL	GBOV	32.9505	-87.3933	Evergreen Needleleaf Forest	2014-2021	294
TUMB	GBOV	-35.6557	148.1513	Evergreen Broadleaf Forest	2019-2020	33
UNDE	GBOV	46.2339	-89.5373	Mixed Forest	2014-2021	217
VALE	GBOV	39.5707	-1.2882	Croplands	2019-2021	33
WOOD	GBOV	47.1282	-99.2414	Grasslands	2014-2021	135

serves as prior knowledge representing the general vegetation state at the coarser scale, rather than direct pixel-to-pixel correspondence. The LightGBM model learns to integrate this coarse-scale information with high-resolution spatiotemporal features (Ln1–Ln11) extracted from similar Landsat pixels, effectively downscaling the coarse prior through local spatial context. This feature integration allows the model to capture fine-scale heterogeneity while benefiting from the temporal stability of GLASS products.

The third set of input features (Ln17–Ln19) consists of environmental variables. We assume that associations between high and low spatial resolution vegetation characteristics depend heavily on environmental contexts. Prior investigations have examined primary environmental drivers controlling temporal vegetation parameter variations, highlighting temperature, water vapor pressure deficit, and solar radiation as dominant factors influencing vegetation parameter temporal dynamics [63]. Therefore, these three environmental parameters were incorporated as final input features for our Landsat LAI reconstruction framework.

Clear-sky Landsat LAI data covered by each training pixel were used as model output.

IV. RESULT

A. Improvement of Fine-Tuning Model

In this section, we selected ten representative GBOV sites for validation, including five forest sites (BART, DELA, JERC, ORNL, and OSBS) and five nonforest sites (BLAN, NIWO, STER, TUMB, and VALE) to assess the efficacy of deep transfer learning in enhancing LAI estimation precision. As shown in Fig. 3, pretrained and fine-tuned DBN models show significant differences across different biomes; fine-tuning significantly improved model performance, especially in the high LAI interval (>4). Taking the BART site as an example, the RMSE of the fine-tuned model decreased from 1.47 to 0.81, and R^2 increased from 0.27 to 0.53, indicating that the model's ability to characterize the high canopy density region has been significantly improved. Similar improvements were also seen at other forest sites, such as DELA (RMSE: 1.67–0.95) and JERC (RMSE: 0.88–0.72). This enhancement primarily results from field-measured observations incorporated during the refinement phase, enabling the model to develop a superior understanding of nonlinear associations between reflectance signatures and actual LAI values, particularly within spectral

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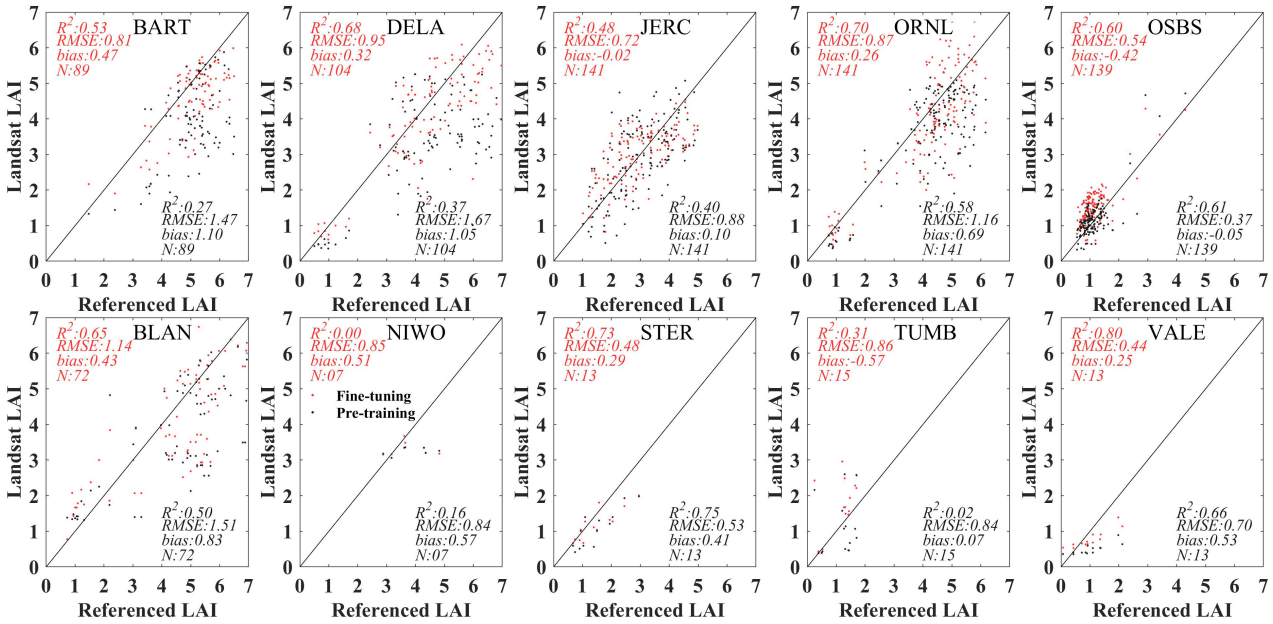

Fig. 3. Scatter plots of comparisons between Landsat clear-sky LAI estimated by pretrained and fine-tuned DBN models at five forest sites (BART, DELA, JERC, ORNL, and OSBS) and five nonforest sites (BLAN, NIWO, STER, TUMB, and VALE).

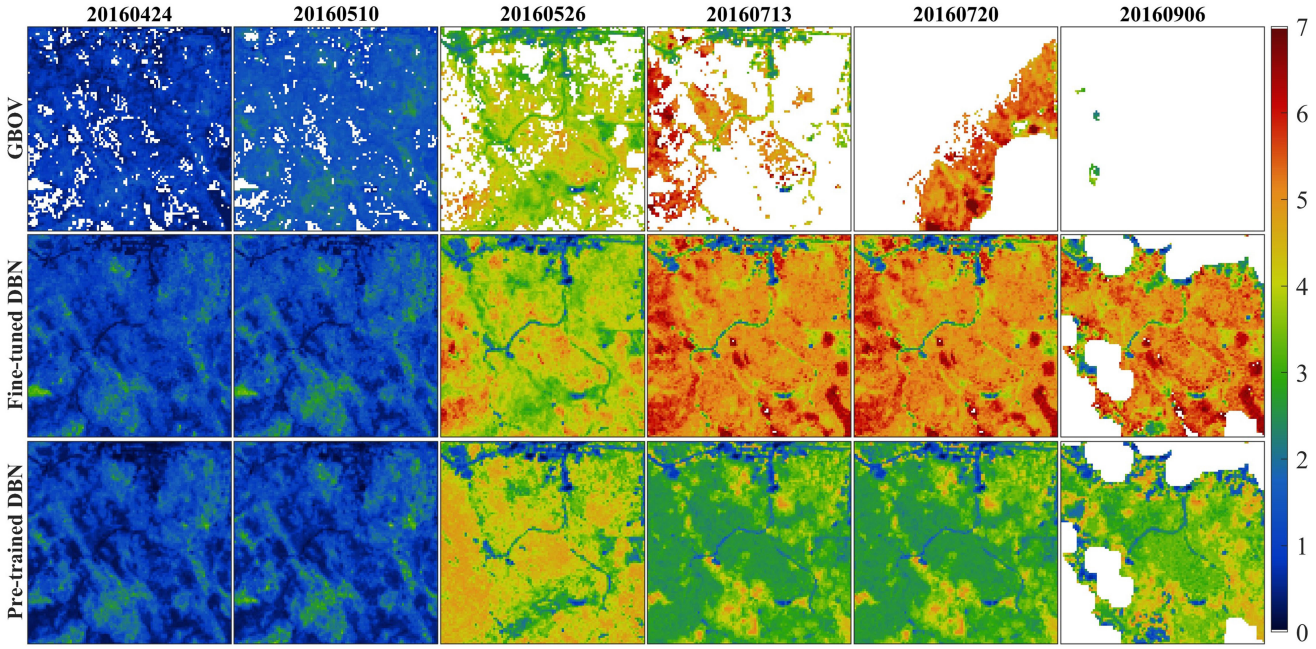

Fig. 4. Visual comparison of GBOV reference, pretrained DBN, and fine-tuned DBN at the BART site in 2016.

saturation domains. For nonforested study areas, discrepancies between the two modeling approaches remain minimal, as the pre-established model demonstrated adequate characterization of low-magnitude LAI ranges. However, it is worth noting that the fine-tuning still resulted in significant improvements at some nonforest sites, such as the RMSE decrease from 1.51 to 1.14 at the BLAN site, suggesting that transfer learning has general significance for improving local-scale LAI estimation.

Fig. 4 presents temporal LAI comparisons among pretrained DBN models, fine-tuned DBN models, and GBOV reference values at the BART site in 2016. Both models capture the seasonal dynamics of vegetation growth, but there are significant differences in numerical accuracy. The pretrained DBN model significantly underestimated LAI magnitudes during peak growing periods (May to July), with a maximum deviation of 1.5, demonstrating the saturation effect of the model trained on satellite products when dealing with high-density vegetation canopy. The fine-tuned DBN model showed better temporal consistency, especially at the peak of the growing season, when the average deviation from the GBOV reference value was reduced to less than 0.5. The fine-tuned model also described the dynamic change characteristics of LAI,

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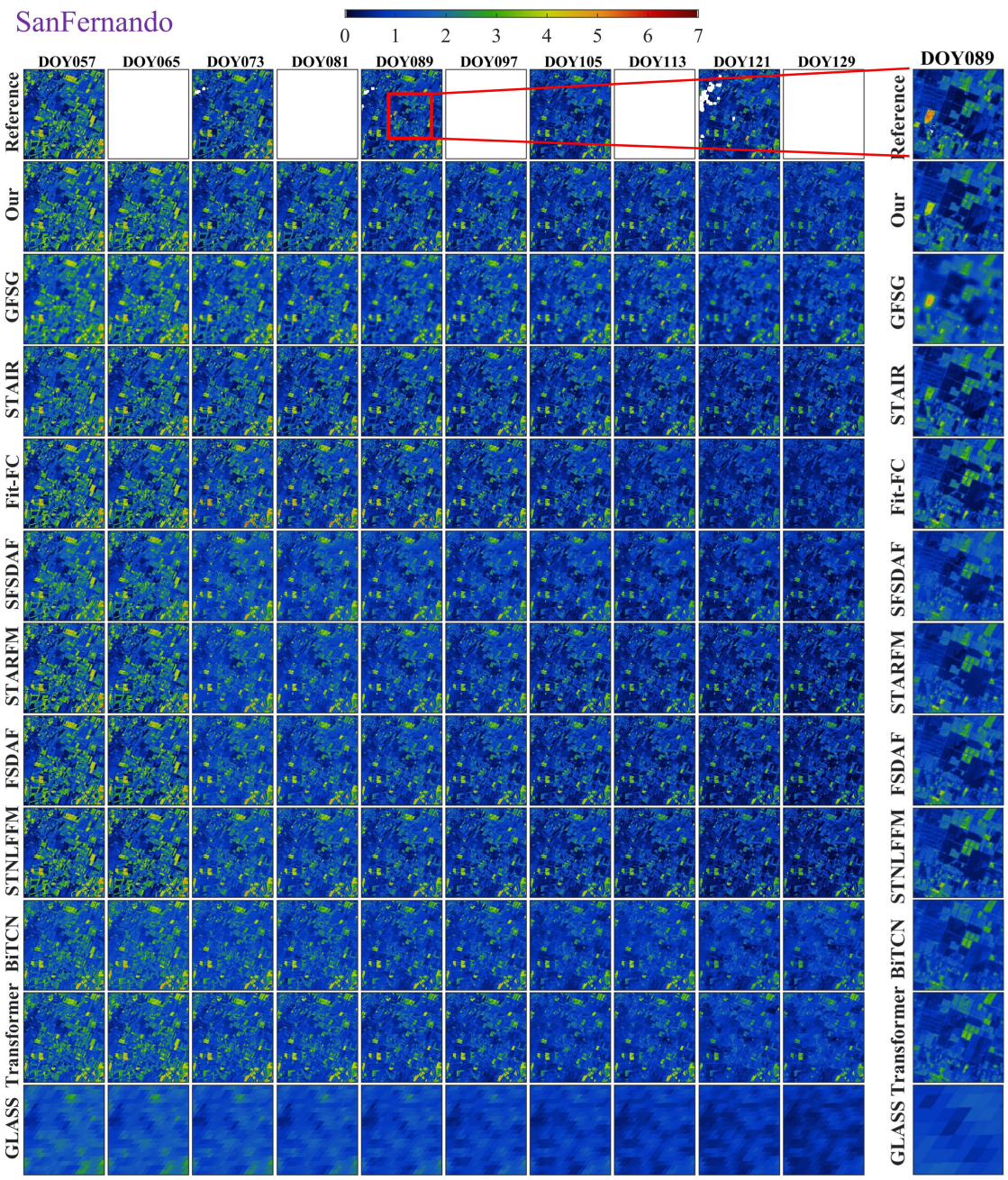

Fig. 5. Visual comparison of reference clear-sky Landsat LAI, our reconstructed LAI, GFSG LAI, STAIR LAI, Fit-FC LAI, SFSDAF LAI, STARFM LAI, FSDAF LAI, STNLFFM LAI, BiTCN LAI, Transformer LAI, and GLASS LAI at the SanFernando sites. The rightmost figures are magnified local views of LAI within the red box.

indicating that the transfer learning strategy effectively improved the adaptability to the seasonal changes of vegetation.

B. Evaluation of the Reconstructed Cloudy-Sky LAI

Figs. 5 and 6 show the spatiotemporal comparison between our method and other reconstructed LAI methods, including GFSG, STAIR, Fit-FC, SFSDAF, STARFM, FSDAF, STNLFFM, BiTCN, Transformer, and GLASS LAI at Pshenichne and SanFernando sites, respectively. These two sites were specifically selected to represent contrasting reconstruction challenges: Pshenichne experiences prolonged cloud cover during the growing season with continuous observation gaps (DOY129-169), representing high cloud-contamination scenarios typical of mid-latitude agricultural regions, while SanFernando exhibits highly heterogeneous agricultural landscapes with fragmented field patterns and diverse crop types, representing challenging conditions for maintaining fine-scale spatial detail during gap-filling. Several reconstruction methods exhibit notable limitations. The GFSG method produces obvious patch effects even with sufficient Landsat clear-sky observations, which is particularly prominent at both sites during periods of rapid vegetation growth. The STAIR method faces challenges in contexts with chronic lack of observational

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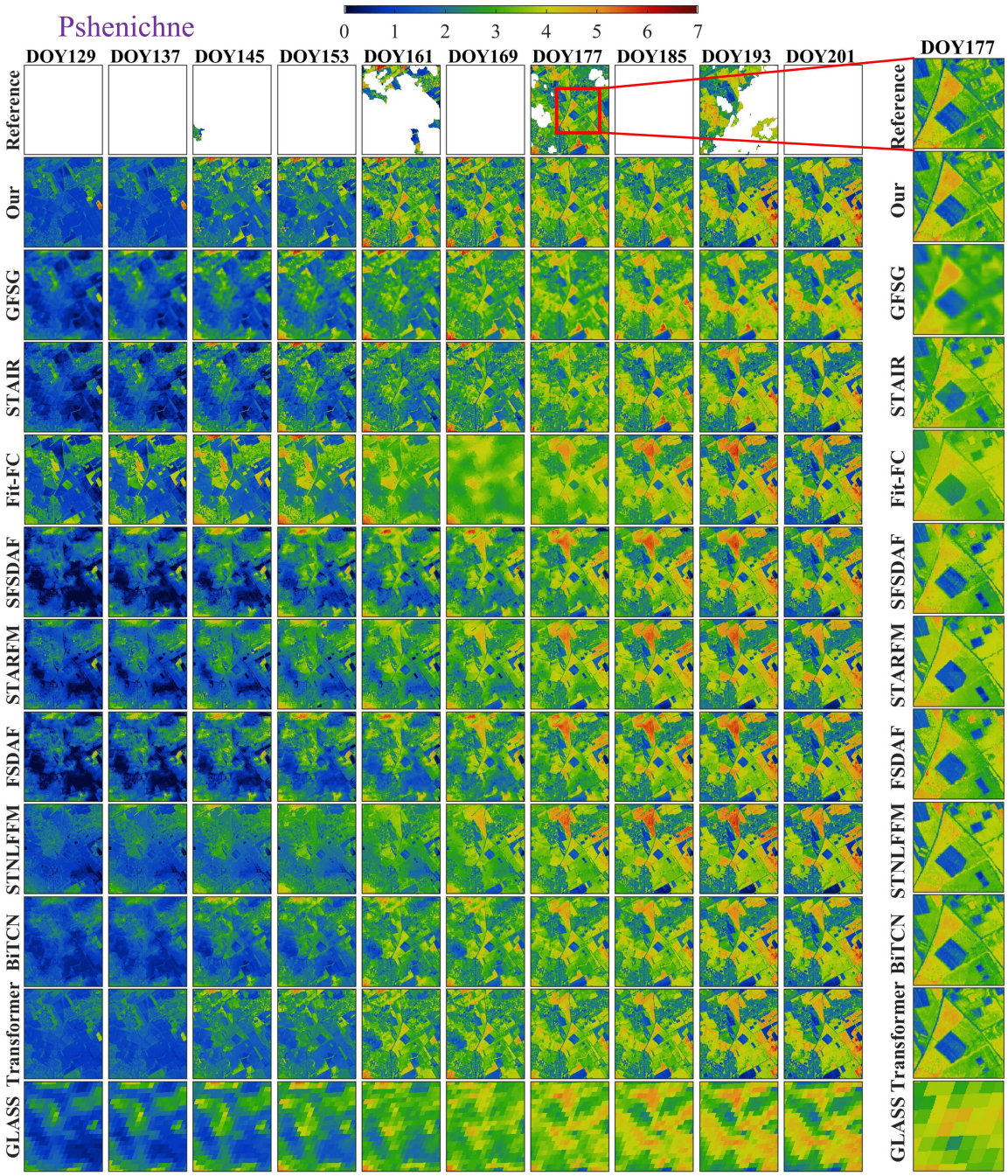

Fig. 6. Visual comparison of reference clear-sky Landsat LAI, our reconstructed LAI, GFSG LAI, STAIR LAI, Fit-FC LAI, SFSDAF LAI, STARFM LAI, FSDAF LAI, FSDAF LAI, STNLFFM LAI, BiTCN LAI, Transformer LAI, and GLASS LAI at the Pshenichne sites. The rightmost figures are magnified local views of LAI within the red box.

data, as evidenced during DOY129-169 at the Pshenichne site and DOY081-121 at the SanFernando site. With few clear-sky Landsat observations available during these periods, LAI reconstructed from STAIR shows significant patching effects away from the observation periods.

The spatiotemporal fusion methods (Fit-FC, SFSDAF, STARFM, FSDAF, and STNLFFM) demonstrate varying degrees of performance limitations. While these methods can maintain some spatial continuity, they often struggle to capture fine-scale heterogeneity and temporal dynamics accurately. STARFM and FSDAF show relatively smoother results but tend to over-smooth spatial details, particularly evident in the agricultural landscape at SanFernando site. SFSDAF and STNLFM exhibit better preservation of spatial patterns but are more susceptible to noise and inconsistencies during periods with limited observations. Fit-FC shows intermediate performance but lacks the robustness needed for consistent reconstruction across different phenological stages. Advanced deep learning methods including BiTCN and Transformer were also evaluated for LAI time-series reconstruction. Both methods exhibit systematic over-smoothing at both sites, resulting in overestimation of low LAI val-

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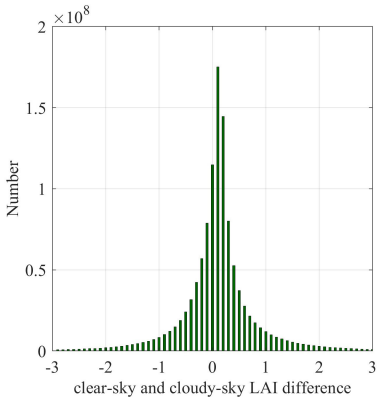

Fig. 7. Histogram of Landsat clear-sky and cloudy-sky LAI difference from 2013 to 2021 over all ImagineS and GBOV sites. The clear-sky LAI values were estimated by fine-tuning the DBN model, and then they were assumed to be covered by clouds; we retrieved the corresponding cloudy-sky LAI by using the reconstruction model.

ues and underestimation of high LAI values. Furthermore, at the Pshenichne site, continuous observation gaps during DOY129-169 severely challenge BiTCN's ability to reproduce high-spatial-resolution LAI.

In contrast, at the SanFernando site, our reconstructed LAI shows a continuous and natural spatial distribution throughout the time series, maintaining clear field boundaries and capturing within-field variability. At the Pshenichne site, even during DOY129-169 when observations were lacking for extended periods, the reconstructed LAI still demonstrates reasonable spatial variation characteristics without significant patch effects. This superior performance is further confirmed by the local magnified areas shown on the rightmost panels of both figures. Our method shows clearer and more accurate spatial details in both cropland (CRO) plot boundaries and LAI distribution within plots compared to all other reconstruction methods. Compared to GLASS LAI products, our 30-m Landsat LAI product can capture obvious fine-scale heterogeneity.

Fig. 7 shows the difference distribution between clear-sky LAI and reconstructed cloudy-sky LAI at all ImagineS and GBOV sites from 2013 to 2021. We assumed that the clear-sky LAI value was a cloud cover condition and used the reconstructed model to derive corresponding cloudy-sky LAI. The histogram shows that the difference between the clear-sky and cloudy-sky LAI has obvious normal distribution characteristics and is concentrated around 0. The statistical results show that more than 85% of the difference falls within the ± 1 range. This compact distribution feature confirms that our proposed LAI gap-filling model can reliably reproduce LAI estimation results under clear-sky conditions.

Fig. 8 presents a comprehensive comparison of the temporal accuracy achieved by various reconstruction methods across multiple sites. At the DELA, HAIN, SCBI, and UNDE locations, our proposed method consistently outperforms both traditional spatiotemporal fusion techniques and state-of-the-art deep learning approaches, achieving an average reduction in RMSE of 0.57. For the STER, TUMB, and VALE sites, all evaluated methods demonstrate comparable performance with minimal accuracy variations. Notably, at the JERC and OSBS sites, although our method exhibits slightly lower precision compared to two specific deep learning methods, it still maintains relatively low RMSE values of 0.38 and 0.65, respectively.

Fig. 9 shows the evaluation results of LAI reconstruction on a Landsat image with path/row No. 128/039 (July 8, 2015). We first assumed all clear-sky pixels of the scene as cloud coverage conditions and used the reconstructed model to obtain the corresponding cloudy-sky LAI values. The comparison results showed that there is a good consistency between reconstructed cloudy-sky LAI and clear-sky reference LAI (R2 = 0.613, RMSE = 0.522), and the density scatter plots [Fig. 9(c)] of the two show that most sample points were concentrated near the 1:1 line, especially in the medium vegetation cover area with LAI values of 2–4. In practical applications, the reconstructed model successfully fills about 30% of the cloud coverage area in the landscape image [the white area in Fig. 9(a)]. The reconstructed results [Fig. 9(b)] maintain the spatial continuity of LAI in the region, and accurately reflect the LAI spatial differentiation characteristics of different vegetation types.

C. Direct Validation of the Landsat 30-m LAI Using Field Data

Fig. 10 presents validation results of LAI estimation using our proposed method across various land cover types. In the comprehensive evaluation of all land surface types [Fig. 10(h)], LAI estimation results under clear-sky and cloudy-sky show good consistency, with RMSE values of 1.06 and 1.11, respectively, confirming that our reconstruction approach successfully preserves LAI estimation precision. Evergreen needleleaf forest (ENF) and evergreen broadleaf forest (EBF) showed relatively good estimation accuracy, achieving RMSE values of 0.89 and 0.79, respectively, under cloudy conditions. However, deciduous broadleaf forest (DBF) and mixed forest (MF) have slightly lower accuracy (RMSE = 1.19–1.42) due to complex seasonal changes and heterogeneous canopy structure, consistent with findings from other LAI estimation research [15]. Notably, although the validation data of grassland (GRA) and shrubland (SH) samples were relatively scarce, high estimation accuracy was achieved under both conditions (R^2 = 0.73–0.76, RMSE = 0.48–0.95).

We validated our method using 18 high-resolution LAI reference maps from ImagineS for different periods. The results showed that the accuracy difference between clear-sky and cloud-sky LAI was generally small, with RMSE ranging from 0.41 to 1.25 and 0.45 to 1.28, respectively, indicating that the transfer learning strategy based on DBN successfully transferred ground measurement knowledge to the field of satellite data. As can be seen from the comparison of spatial distribution above Fig. 11, the consistency between our estimated LAI and ImagineS reference LAI can be summarized in three aspects. First, the grasp of the LAI range was relatively accurate, which was most prominent at 25MAYO and Barrax sites, where alfalfa was planted in rotation, and the change of LAI value from bare soil to dense vegetation was well depicted. Second, most of the ImagineS sites are farmland surfaces with strong heterogeneity, and this method still maintains a good ability to express spatial details, which

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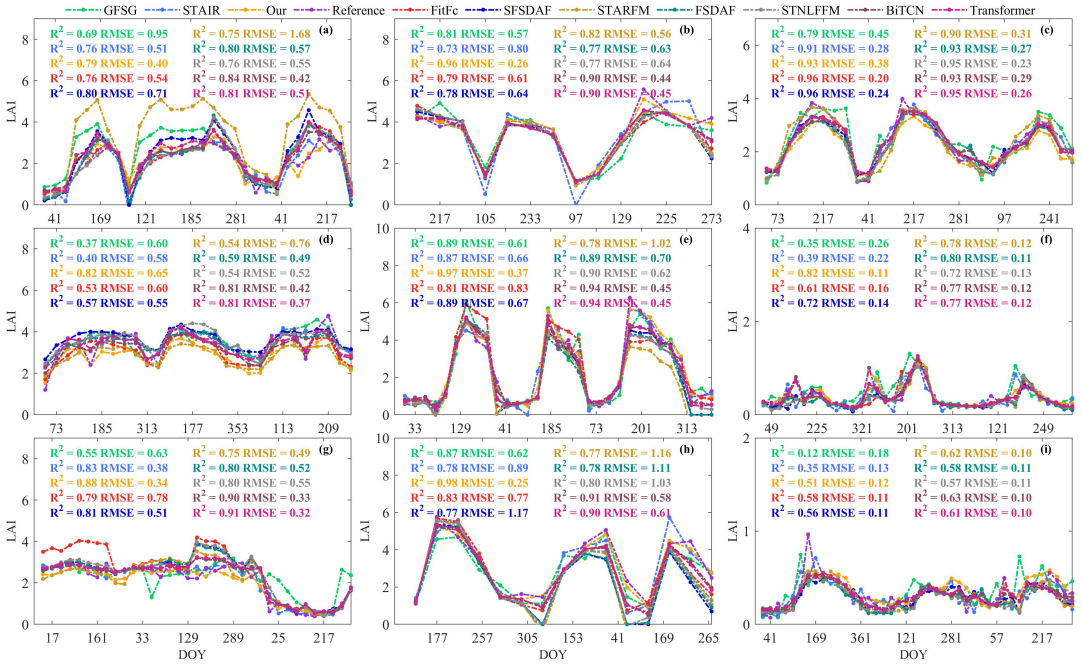

Fig. 8. Comparisons of GFSG LAI, STAIR LAI, our reconstructed LAI, the actual reference LAI, FitFc LAI, SFSDAF LAI, STARFM LAI, FSDAF LAI, STNLFFM LAI, BiTCN LAI, and Transformer LAI from 2018 to 2020 at nine GBOV sites. (a) DELA. (b) HAIN. (c) JERC. (d) OSBS. (e) SCBI. (f) STER. (g) TUMB. (h) UNDE. (i) VALE.

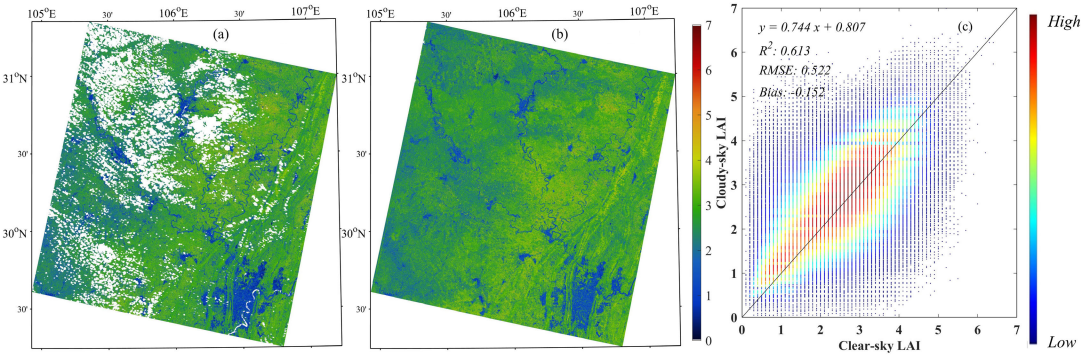

Fig. 9. Comparison of (a) Landsat clear-sky LAI and (b) Landsat cloudy-sky reconstructed LAI for path/row 128/039 on July 8, 2015. (c) Density scatter plot and statistics between them; the color from blue to red represents the density of points from low to high.

indicates the adaptability of transfer learning strategies under complex surface conditions. Third, the numerical accuracy and spatial details of the reconstruction results under the cloud are comparable to those of the clear-sky estimation, which is mainly due to the full use of the spatiotemporal information of local similar pixels. In addition, from the perspective of the temporal dimension, these LAI maps showed stable performance in different phenological periods (May–July) and years (2013–2015), which further confirmed the reliability of the proposed method.

D. Spatial and Temporal Patterns of Landsat LAI

Fig. 12 shows the spatiotemporal continuous changes of Landsat LAI at SouthWest sites in France in 2015. During DOY001-073, LAI values of most pixels in the study area were relatively low (<2), and only some fields planted with overwintering wheat maintained high values (2 and 3), which reflected the slow growth stage of winter wheat after overwintering. During the DOY081-137 period after spring, we observed a significant increasing trend in LAI of large area pixels, which mainly reflected that winter wheat entered a rapid growth period and LAI values continued to rise until DOY129 reached its growth peak (LAI > 4). Subsequently, during DOY145-161, these areas with high LAI values suddenly disappeared, indicating that winter wheat entered the harvest period and the farmland landscape underwent drastic changes. During DOY161-249, LAI of some farmland pixels began to increase significantly again, which reflected the vigorous growth period of spring sowing crops (mainly corn and sunflower) and formed a new region with high LAI values.

From DOY257–297, these higher LAI values showed a gradually decreasing trend, corresponding to the maturity and harvest stages of corn and sunflower. At the end of the growing season (DOY305-361), new vegetation signals appeared again in some fields, which may reflect a new round of winter wheat

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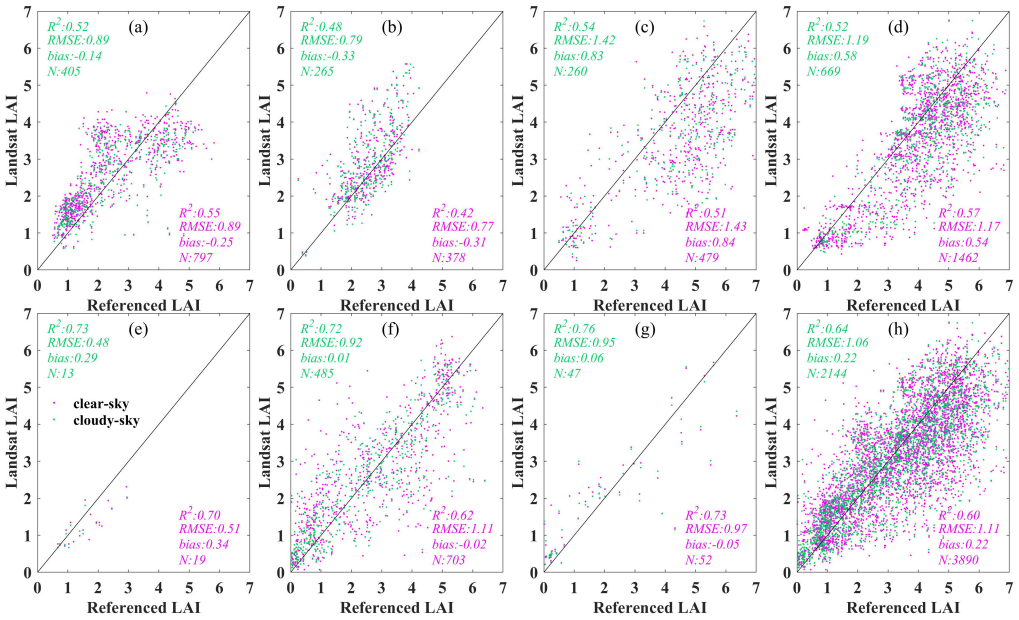

Fig. 10. Validation scatterplots of Landsat clear-sky and cloudy-sky LAI using the GBOV and ImagineS 30-m in situ LAI measurements at different land cover types (a) ENF, (b) EBF, (c) DBF, (d) MF, (e) GRA, (f) CRO, (g) SH, and (h) all land cover types.

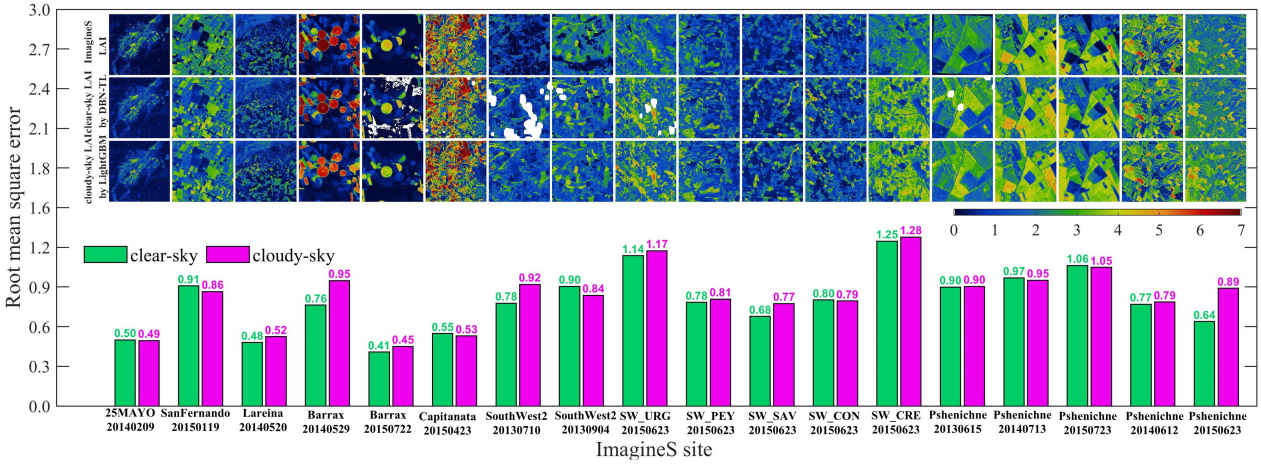

Fig. 11. RMSE distribution map for validating Landsat clear-sky LAI and cloudy-sky LAI using ImagineS reference LAI maps. The ImagineS reference LAI, clear-sky LAI, and cloudy-sky LAI at different sites are also shown.

sowing and emergence process. The whole sequence clearly demonstrated the complex agricultural landscape dynamics in the study area. Our method successfully captured the LAI change characteristics of different crops during key phenological periods such as growth and harvest, confirming its effectiveness in describing the spatiotemporal dynamics of farmland ecosystems.

Fig. 13 shows spatial distributions of Landsat, GLASS, and MODIS LAI products in China on the 1st and 225th days of 2017. The most obvious defect of MODIS LAI products [Fig. 13(e) and (f)] is the presence of a large number of missing values (white areas), especially in the snow-covered areas of the northwest desert and plateau. In addition, in cloudy areas such as southwest China, the MODIS main algorithm relies on high-quality reflectance and fails when encountering cloud and fog interference, resulting in a significant underestimation of LAI values. GLASS LAI products [Fig. 13(c) and (d)] are generated by using a time-series deep learning model, which has strong resistance to cloud and fog interference, so the spatial coverage is more complete, especially in the southwest cloudy area, showing better spatial continuity. GLASS LAI is the basic data for training the Landsat LAI estimation model, so Landsat LAI products [Fig. 13(a) and (b)] inherit the complete spatial coverage advantage of GLASS LAI, and the two products are also consistent in macro spatial distribution. Due to their higher spatial resolution, in Fig. 13(b) on day 225, especially in mountainous areas and areas with complex agricultural landscapes, Landsat LAI could capture more detailed LAI spatial variation.

Fig. 14 shows the distribution of Landsat, GLASS, and MODIS LAI products along latitudinal gradients on the first and 225th days of 2017. In winter [Fig. 14(a)], the three

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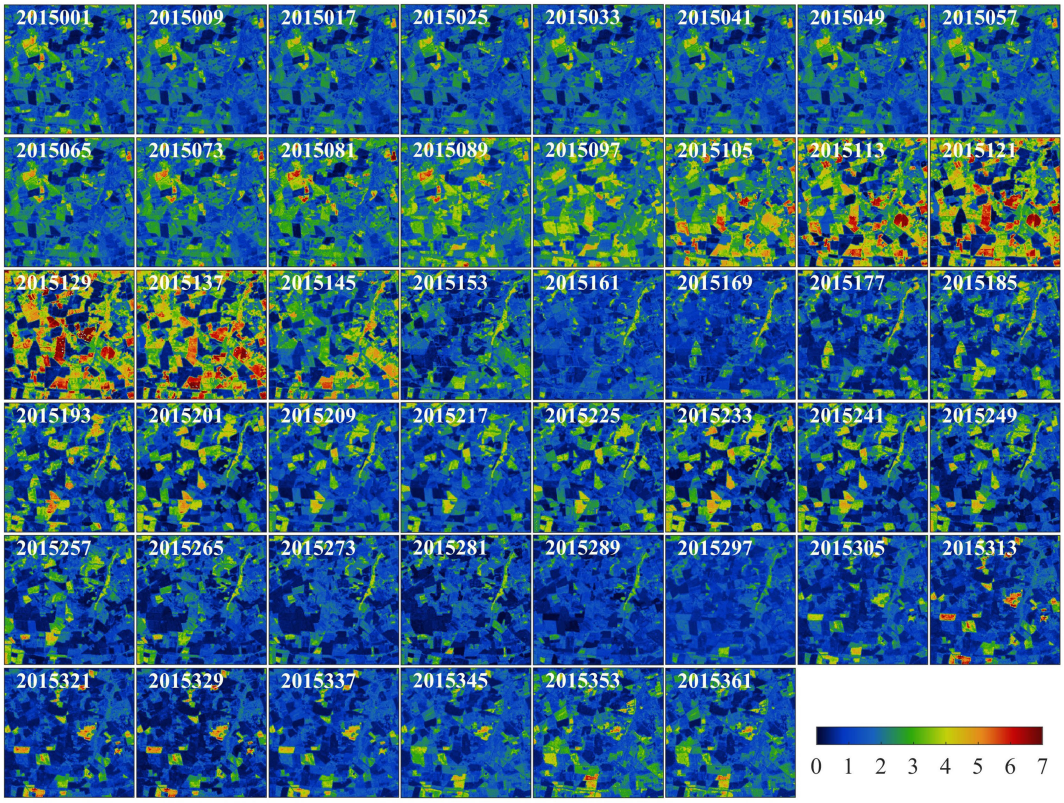

Fig. 12. Spatial pattern of the estimated Landsat LAI at the SouthWest site in 2015.

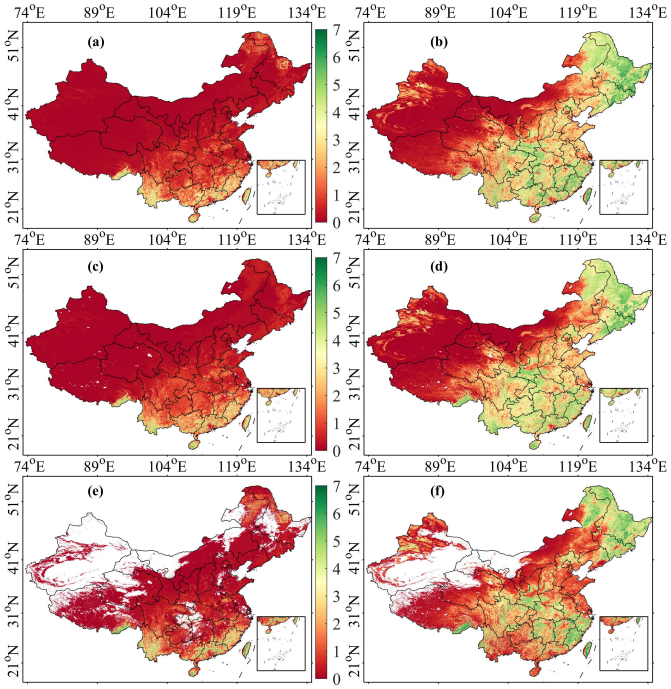

Fig. 13. Spatial distributions of Landsat, GLASS, and MODIS LAI products in China on the 1st and 225th days of 2017. (a) Landsat 2017001. (b) Landsat 2017225. (c) GLASS 2017001. (d) GLASS 2017225. (e) MODIS 2017001. (f) MODIS 2017225.

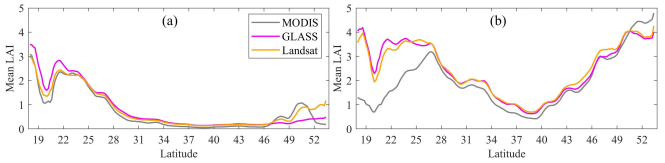

Fig. 14. Comparison of Landsat, GLASS and MODIS LAI products based on latitude gradient in China (a) on the first day of 2017 and (b) on the 225th day of 2017.

products all show higher LAI values in low-latitude regions (19°N–25°N), reflecting the high vegetation vitality maintained throughout the year in tropical and subtropical regions. LAI values generally decline with increasing latitude, which is consistent with winter defoliation and dormancy in the Northern Hemisphere. It is worth noting that MODIS products are significantly lower than the other two products in some latitude ranges, such as near 20°N, which is related to cloud pollution and algorithmic limitations. The summer growing season [Fig. 14(b)] shows a completely different distribution pattern, and LAI values increase significantly in the middle and high latitudes (35°N–50°N), and even exceed those in the northern temperate forest region. Landsat and GLASS products have similar overall trends but different details, while MODIS products are systematically deviated from the other two, especially at low latitudes.

Fig. 15 shows the LAI time-series changes of different biome sites from 2014 to 2021. The Landsat LAI sequences generated in this study exhibit complete time series and successfully capture the seasonal dynamic characteristics of different vegetation types. Notably, despite GLASS LAI serv-

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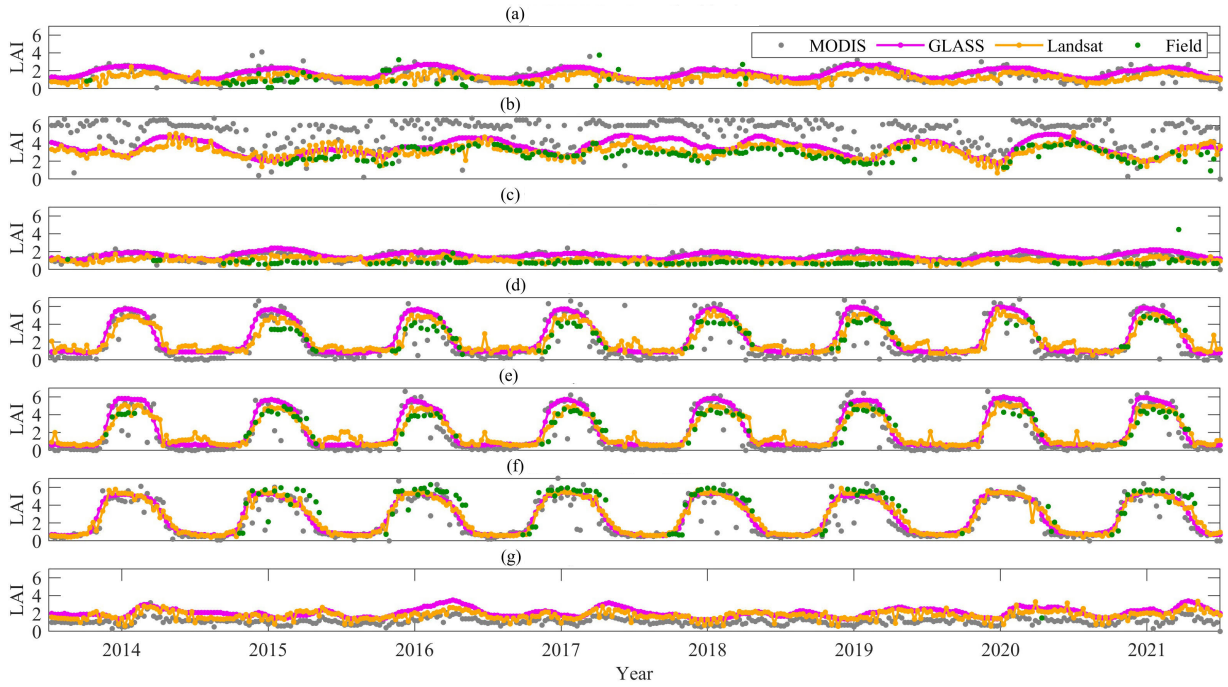

Fig. 15. Time series of the Landsat LAI and GLASS and field LAI values from 2014 to 2021 over (a) DSNY, (b) GUAN, (c) OSBS, (d) STEI, (e) UNDE, (f) SERC, and (g) LAJA sites.

ing as an output variable for the pretrained model, Landsat LAI shows significant differences from both GLASS and MODIS LAI at multiple sites, such as OSBS ENF, STEI DBF, and UNDE MF; this divergence is mainly caused by differences in spatial resolution. GLASS and MODIS LAI represent the average of 250–500-m mixed pixels, while Landsat LAI with 30-m resolution is better able to characterize pure pixel features, and therefore exhibits greater consistency with ground measurements. MODIS LAI displays considerably more anomalous values, primarily resulting from the use of backup algorithms when the main RTM fails to converge. In contrast, our Landsat LAI and GLASS LAI demonstrate smoother time series with more consistent seasonal patterns. This smoother temporal profile is particularly evident at deciduous and MF sites (STEI and UNDE), where phenological transitions are captured with greater fidelity. In addition, short-term fluctuations in LAI time series at some sites (such as GUAN and STEI sites) are mainly caused by misidentification of clouds and cloud shadows, this fluctuation can be eliminated by improving the accuracy of cloud detection or introducing a time-series filtering method.

V. Discussion

A. Comparison With Related Deep Transfer Learning Approaches

Recent studies have explored deep transfer learning for high-resolution LAI estimation. Notably, Zhou et al. [35] proposed a Bi-LSTM framework pretrained on MODIS reflectance and LAI time series, fine-tuned with U.S. field measurements, and reconstructed Landsat imagery via spatiotemporal fusion. While sharing the conceptual foundation of transfer learning, our study presents fundamental methodological distinctions that address critical limitations. First,

TABLE III
Accuracy Comparison of High-Resolution LAI Estimation Methods

Reference	Methods	Report accuracy (RMSE)
Zhou et al. (2023) [35]	Deep transfer learning	0.37-1.51
Jin et al. (2024) [15]	Deep learning	0.95
Ma et al. (2025) [64]	Deep learning	0.985
Our	Deep transfer learning	0.44-1.14

Bi-LSTM requires complete time series as input/output, yet MODIS products contain cloud-induced gaps and temporal noise. Their reliance on MODIS LAI, which exhibits documented accuracy limitations and temporal fluctuations, risks learning from anomalous values. Our DBN employs point-to-point mapping using cloud-free MODIS reflectance paired with superior-quality GLASS V6 LAI, ensuring robust pretraining without temporal continuity requirements. Second, their fine-tuning used nine U.S. sites, potentially limiting generalizability, whereas our 40 globally distributed ImagineS and GBOV sites spanning seven biomes enhance cross-regional applicability. Third, and most critically, Zhou et al. [35] relied on existing spatiotemporal fusion methods (FSDAF/STAIR) for reflectance reconstruction—approaches with known limitations in heterogeneous landscapes and failure when both MODIS/Landsat observations are unavailable. We introduce a novel LightGBM-based framework operating directly in LAI space, integrating 19 features including spatiotemporal autocorrelation, environmental drivers, and phenological intelligence that significantly outperforms seven benchmark methods. Finally, our continental-scale China product (2013–2021) demonstrates operational scalability beyond

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regional validation. These distinctions constitute substantive innovations in pretraining quality, gap-filling methodology, and global applicability. To show the performance of our method, we compared validation accuracies with recently published high-resolution LAI products (Table III). Zhou et al. [35] reported RMSE of 0.37–1.51 using deep transfer learning across U.S. CROs, GRAs, and forests. Jin et al. [15] achieved RMSE of 0.95 using deep transfer learning approaches, while Ma et al. [64] demonstrated RMSE of 0.98 with time-series deep learning methods. Our fine-tuned DBN model achieves comparable or superior accuracy; these confirm that our framework delivers competitive accuracy.

B. Effectiveness of Deep Transfer Learning in Improving Estimation of Landsat LAI

The proposed deep transfer learning strategy significantly improved the estimation accuracy of Landsat LAI, especially in the high-value LAI interval (for example, RMSE of forest plots decreased from 1.47–1.67 to 0.81–0.95). This performance improvement is mainly due to the unique knowledge transfer mechanism of deep transfer learning. Traditional deep learning models often require a large amount of labeled data to obtain good performance [65], [66], while in the field of LAI inversion, it is extremely difficult and costly to obtain sufficient ground-measured data. Relying on a single data source for training, such as using only satellite products or simulation data, will make it difficult for the model to adapt to LAI characteristics at different scales. For example, the pretraining model was only trained on GLASS LAI with a resolution of 250 m, which could not fully express the spatial heterogeneity of LAI at a resolution of 30 m, resulting in significant underestimation of high-value LAI. By learning feature representations in the source domain (GLASS LAI), deep transfer learning can effectively migrate to the target domain (ground-measured LAI), thus reducing the need for labeled data [53].

The effectiveness of our adopted pretraining and fine-tuning strategy lies primarily in its hierarchical feature learning and selective parameter updating mechanism. In the pretraining stage, DBN can establish a feature hierarchy from simple to complex through layer-by-layer greedy learning [67], which enables the model to capture the complex nonlinear relationship between LAI and reflectance. In the fine-tuning stage, the strategy of freezing the feature extraction layer and updating only the high-level parameters can effectively prevent overfitting and maintain the generalization ability of the model [68]. The success of this strategy is consistent with the latest theory of neural network optimization, that is, low-level features have stronger generality, while high-level features are more suitable for specific tasks [69]. By introducing a transfer learning strategy, our method not only improves the numerical accuracy but, more importantly, solves the saturation effect problem of LAI with high values. At sites where farmland landscapes are fragmented, such as SanFernando and Pshenichne, the fine-tuned model successfully maintains fine spatial heterogeneity.

The success of transfer learning in bridging the domain gap between coarse-resolution satellite products and fine-resolution ground measurements can be attributed to hierarchical feature learning: the pretraining stage captures universal spectral-LAI relationships governed by radiative transfer principles, while fine-tuning adapts these representations to local-scale heterogeneity and corrects systematic satellite-ground biases. Fig. 3 provides direct experimental evidence: fine-tuned models show substantial accuracy improvements over pretrained models across ten GBOV sites, with the most pronounced gains in high-LAI forests where pretrained models exhibit severe saturation. This demonstrates that transfer learning effectively overcomes the spectral saturation problem inherent in satellite-only training by incorporating ground-truth knowledge of high-density canopies.

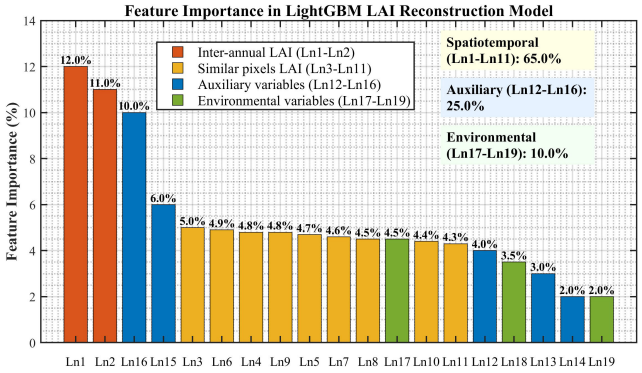

Fig. 16. Relative importance of 19 input features in the LightGBM LAI reconstruction model.

C. Advantages of Spatiotemporal Constraints in LAI Reconstruction

In the cloudy and foggy area (e.g., Pshenichne site DOY129-169), the reconstructed LAI maintains a good spatiotemporal continuity despite a long-term lack of effective observation, while the existing methods tend to show a significant patchy effect far from the observation period. This performance advantage is mainly due to the comprehensive utilization mechanism of multisource spatiotemporal information. According to the first law of geography proposed by [70], similar regions tend to have similar characteristics. Based on this theory, we adopted the strategy of selecting similar pixels with a 3×3-km spatial window to make full use of the spatial autocorrelation characteristics of vegetation growth. Similar studies have also shown that by calculating the correlation coefficient of seasonal curves to identify similar pixels, spatial distance and phenological similarity can be considered at the same time, and this careful consideration of spatial and phenological characteristics can significantly improve the reconstruction accuracy [27]. To quantify the contribution of different feature categories, we analyzed the relative importance of the 19 input features (Fig. 16). Results show that spatiotemporal information (Ln1–Ln11) dominates with 65% total importance, wherein interannual LAI values (Ln1 and Ln2: 23%) and similar-pixel LAI values (Ln3–Ln11: 42%) provide the strongest constraints. Auxiliary variables (Ln12–Ln16) contribute 25%, with GLASS LAI (10%) being the most influential coarse-resolution prior. Environmental variables (Ln17–Ln19) account for 10%, with temperature

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(4.5%) as the primary climatic driver. This hierarchy confirms that temporal autocorrelation and spatial phenological similarity constitute the dominant gap-filling mechanisms, while environmental constraints and coarse-resolution priors provide essential complementary information.

Another key innovation in the reconstructed model is the introduction of environmental variables as physical constraints. According to the study of [63], environmental factors such as temperature, water vapor pressure deficit, and solar radiation are closely related to the vegetation growth process, which provides a theoretical basis for us to choose the driving variables of the reconstructed model. Different from the traditional pure statistical interpolation method, we integrate these key environmental variables to constrain the reconstruction process, and the introduction of environmental variables can effectively improve the physical rationality of the reconstruction results [71]. In addition, the robustness of the model can be improved by constructing the seasonal curve of vegetation growth using the time-series information of many years [72]. This reconstruction strategy based on multisource constraints can significantly improve the reliability of the results.

D. Limitations and Future Improvements

Several constraints exist within this study. The first is the representation of training data. While ImagineS and GBOV networks provide high-quality in situ measurements, some ecosystem types are still under-represented; for example, GRA (LAJA) and shrub (ONAQ) sites have small sample sizes ($N < 100$), which may limit the ability of the model to generalize to these vegetation types. Deep learning model performance depends heavily on training sample quantity and representativeness, especially in the inversion of vegetation parameters, where the structure and physiological characteristics of different ecosystems differ significantly; sufficient samples are needed to capture this variability [33]. In the future, data from other observation networks such as VALERI and Bigfoot can be integrated to expand the ecosystem coverage of training samples, thereby improving model generalization capabilities. Our validation demonstrates that the fine-tuned model effectively mitigates high-LAI underestimation across all available ground measurements (maximum LAI ~7 at forest sites), with systematic bias substantially reduced in dense canopies (Figs. 3 and 4). However, model performance in extreme high-LAI regions ($LAI > 7$) such as tropical rainforests remains untested due to the absence of validation data from these ecosystems. Extending the framework to such regions would require incorporating field measurements from tropical forests to further calibrate the model for extreme canopy densities.

The second limitation concerns the fixed 3×3-km window for similar pixel selection in the LightGBM reconstruction model. Our sensitivity analysis (Section III-C) showed that this window size achieves optimal balance between reconstruction accuracy and computational efficiency across validation sites. However, the fixed window may not be equally suitable for all ecosystems—smaller windows might suffice for homogeneous forests with consistent phenology, while larger windows may be needed in fragmented agricultural landscapes with high-spatial heterogeneity. Despite using a uniform window size, our validation across diverse ecosystems (Fig. 10) demonstrates satisfactory performance (RMSE = 1.06–1.42), suggesting the 3-km window provides reasonable adaptability. Nevertheless, an adaptive window selection strategy that adjusts spatial extent based on landscape heterogeneity could further optimize reconstruction performance and represents a direction for future methodological refinement.

The third limitation involves data source diversity. Currently using only Landsat data can be challenging in cloudy and foggy areas, especially in areas like Chengdu, China, where the scarcity of valid observations can degrade the performance of the reconstructed model. Cloud contamination frequency varies substantially across China (>70% in southwestern regions versus <30% in northwestern areas), directly impacting reconstruction reliability. Future operational products should incorporate quality flags indicating the number of clear observations used. In high cloud-cover regions, single-sensor observation frequency often fails to support continuous monitoring requirements [73]. Observation frequency could be enhanced by integrating multisource high-resolution satellite observations from Sentinel-2, Landsat 9, and GF-6 to improve temporal data continuity, thereby strengthening reconstruction reliability. Regarding computational scalability, our tile-based processing framework is well-suited for distributed computing on cloud platforms. Processing a single Landsat tile for one year requires approximately 2.5 h on a standard workstation (Intel Xeon, 32 GB RAM). For continental-scale applications, parallel processing on cloud platforms can complete annual LAI generation for China within three to five days using 100 computing nodes, demonstrating the feasibility of global-scale operational production with reasonable computational costs.

The fourth limitation concerns multiscale input integration in the LightGBM model. While integrating features across three spatial scales (30-m Landsat LAI, 250-m GLASS LAI, 0.25° ERA5 variables) enables robust reconstruction, scale mismatches may introduce errors in highly heterogeneous landscapes. Our validation at heterogeneous agricultural sites (Figs. 5 and 6, e.g., SanFernando with fragmented fields <100 m) shows that the method maintains fine-scale spatial detail, suggesting that high-resolution spatiotemporal features (Ln1–Ln11) effectively compensate for coarse-resolution inputs. However, we acknowledge that in extremely heterogeneous or rapidly changing areas, coarse-resolution priors may not fully capture micro-scale variability, and uncertainties from multiscale integration are not explicitly quantified. Future work should incorporate scale-aware uncertainty estimation frameworks and adaptive weighting schemes that adjust feature contributions based on local heterogeneity to further improve reconstruction reliability in complex landscapes.

The last is the effect of terrain. At the 30-m scale, topography significantly affects the estimation accuracy of vegetation parameters by changing the solar incidence angle and shadow effect. This effect is particularly pronounced in mountainous areas and may lead to systematic bias in LAI estimates. In future studies, the application of terrain correction models such as path length correction (PLC) in remote sensing reflectance

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preprocessing is expected to further improve the inversion accuracy of LAI in mountainous areas [74].

VI. Conclusion

This study proposes a new framework for generating a 30 m, eight-day LAI product in China spanning 2013–2021 from Landsat data using deep transfer learning. Our approach addresses two key challenges in high-resolution LAI estimation: the difference between satellite-acquired LAI and ground-measured LAI, and the discontinuity in the time series due to cloud contamination. The framework uses a three-step approach, including pretraining DBN models using MODIS surface reflectance and GLASS LAI, fine-tuning using ground measurements, and reconstruction of cloudy-sky LAI using spatiotemporal constraints and environmental variables.

Comprehensive validation using global in situ measurements demonstrated the effectiveness of our approach, with overall RMSE values of 1.06 and 1.11 for LAI on clear and cloudy days, respectively. The fine-tuning process significantly improved model performance, especially in high LAI areas such as forests, reducing RMSE from 1.47–1.67 to 0.81–0.95. The reconstructed model successfully maintains the spatiotemporal continuity of the LAI time series, even over longer cloud cover times, and is superior to existing methods such as GFSG and STAIR, Fit-FC, FSDAF, SFSDAF, STARFM, STNLFFM, BiTCN, and Transformer in preserving spatial detail and avoiding plaque effects.

This study contributes an innovative deep transfer learning technique for generating high spatiotemporal resolution LAI. This data is of great value for precision agriculture, ecosystem modeling, and climate change research. Although this study has validated the method in multiple ecosystems around the world, future work will further increase the representation of training samples, integrate multisource satellite data to increase observation frequency, and introduce terrain correction models to improve estimation accuracy in mountainous areas. The final goal is to build globally consistent high spatiotemporal LAI products and provide a more solid data foundation for Earth system science research.

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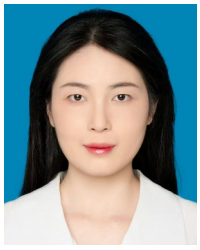

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